

- 7 A transformer has 3000 turns in its primary coil. It is used to convert a main supply of root-mean-square value of 250 V to an alternating voltage having a peak value of 14 V.

(a) Explain what is meant by the *root-mean-square* value of an alternating current.

.....
 [2]

(b) Calculate the number of turns in the secondary coil. State an assumption made in your calculation.

number of turns = [2]

Assumption:
 [1]

(c) The secondary coil is connected in series with a resistor R. The variation with time t , in seconds, of the potential difference in the secondary coil is given by the expression

$$V = 14 \sin(475t)$$

(i) To prevent overheating, the mean power dissipated in R must not exceed 340 W.

Calculate the minimum resistance of R.

resistance = Ω [2]

(ii) On Fig. 7.1, sketch the variation with time t of the power P dissipated in R for a time interval of one period. Numerical values are not required.

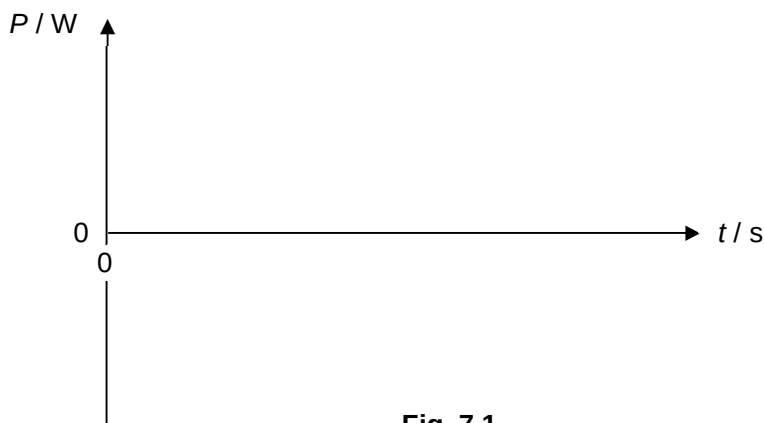


Fig. 7.1

[2]

8 Read the passage below and answer the questions that follows